

Bottom-up derivation
From quantum decoherence physics

Joos & Zeh (1985): gravitational decoherence

$\tau_D \sim \hbar R_{\text{vir}}/GM^2 \approx 10^{-70} \text{ s for halos}$
instantaneous on cosmological timescales

↓ *rate-limiting step
is virialization at H^{-1}*

Press-Schechter halo mass function

$$\dot{I} = \int \frac{dn}{dM} \cdot \frac{M}{m_p} \cdot H(a) \cdot \frac{d \ln F_{\text{coll}}}{d \ln a} dM$$

↓ *change variables to $v = \delta_c/(\sigma D)$*

Peak height integral

$$\dot{I} \propto \rho_m \cdot f \cdot H \cdot D^2 \int_0^\infty v^2 e^{-v^2/2} dv$$

integral = $\sqrt{\pi/2}$ + $\langle v \rangle_{\text{eff}} \propto D^{1/2}$ at nonlinear scale

↓

Bottom-up result

$$\dot{I} \propto \rho_m(a) \cdot D(a)^{5/2} \cdot f(a) \cdot H(a)$$

$$n = 5/2$$

Top-down derivation
From horizon thermodynamic consistency

Required activation function

$\mathcal{E}(a) = \exp(1 - 1/a)$ from Jacobson-Cai-Kim
requires $S_{\text{info}}(a) \propto -1/a$

↓ *matter-domination
scalings*

Matter-domination scalings

$$H \propto a^{-3/2}, \quad A_H \propto a^3, \quad T_H \propto a^{-3/2}, \quad D \propto a$$

↓ *substitute into
entropy integral*

Entropy integral scaling

$$dS_{\text{info}}/d \ln a \propto a^{n-9/2}$$

require result $\propto a^{-1}$ to reproduce $\mathcal{E}(a) \propto \exp(-1/a)$

↓

Top-down result

$$n - 9/2 = -1 \quad \Rightarrow \quad n = 5/2$$

$$n = 5/2$$

Both directions yield $n = 5/2$. Agreement constitutes a derivation, not a scaling argument.